



Document prepared for WSB, Poznań

Web application programming

Assessment requirements and grading system.

Computer Science, 1st Cycle Degree of Studies

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Introduction

Passing the class is based on the following components:

- solving the small assignments published on the Moodle platform: 0 – 20 points,
- creating project of a web application according to the design guidelines contained in this document: 0 – 90 points,
- demonstrating the project: 0 – 10 points.

Note: the demonstration can be replaced with multimedia material or extended documentation, showing the compliance of the solution with the criteria described in this document (when choosing project of a bigger application) or published on the Moodle platform (when choosing smaller tasks).

A score ranging from 0 to 100 points is used along with the WSB Poznań grading scale:

- fail (0 – 49 points),
- satisfactory (50 – 59 points),
- more than satisfactory but less than good (60 – 69 points),
- good (70 – 79 points),
- very good (80 – 89 points),
- excellent (90 – 100 points).

If the total sum of 100 points obtained for the course is exceeded, the final result will be rounded down to a maximum of 100 points.

This document, in the next part, tries to answer the questions that may arise in connection with the implementation of the above-mentioned activities.

The number of participants in Project Teams

It is allowed to carry out activities both individually and in a team form. In the case of a team implementation, it is advisable that the size of such a team does not exceed four People. In exceptional cases, groups with a larger number of people are allowed, but it requires individual arrangements with the Lecturer.

Project technology

Among the recommended technologies for the implementation of the project, the ones discussed during the course should be distinguished, while it is permissible to use any technology that uses widely understood scoring mechanisms described in the sections below on project evaluation criteria.

The form of providing the solutions

The recommended form of providing project sources is the Moodle platform through the activities within the course created for this purpose and by providing access to the Git version control system where project sources are located. However, all other forms of providing project sources are allowed, as a result of which the Lecturer will receive them in a readable and compliant with generally accepted standards form.

Sample project topics

Below is a list of exemplary project topics – it is allowed to modify the proposed topic or present the project's own topic by the Project Team, and, after approval by the Lecturer, its implementation. The only requirements in this case are the compliance of the project topic with the scope of the course and generally accepted standards.

1. Task List application
2. Training planner application
3. Survey system
4. Application for managing and sharing photos
5. Application to support learning a foreign language
6. An application supporting healthy eating
7. Shopping list application
8. Application for managing the restaurant menu
9. Knowledge base application
10. Application supporting the hotel reservation system
11. Phone book application with a phone call simulator
12. Playing Tic Tac Toe (without AI)
13. Application for remote monitoring of machine parameters (e.g. disk usage, memory load)
14. Application supporting the inventory of comic books, books or CDs
15. Application supporting the doctor's registration system
16. Virtual Auction Application

17. Application supporting the management and presentation of the product catalog
18. Electronic journal of grades

Suggested time limits

Suggested time limit for taking activities ends with the beginning of the last classes in the semester, where projects are planned to be demonstrated by Project Teams.

Project evaluation criteria

The project will be assessed according to the following criteria.

1. Specification: 0 – 14 points, including:
 - a. defining a short description of the project and the problem to be solved by the designed application: 0 – 2 points,
 - b. indication of at least four arguments why the web application technology was selected as part of the project to solve the assumed problem or indication, with at least four arguments, why the web application technology will not work for a given problem: 0 – 4 points, where single argument = 1 point,
 - c. determination of at least four functional requirements: 0 – 2 points, where single requirement = 0.5 point,
 - d. determination of at least two non-functional requirements: 0 – 2 points, where single requirement = 1 point,

- e. indication of at least two potential recipients of the system: 0 - 2 points, where 1 recipient = 1 point,
 - f. indication of at least two potential goals for creating a system: 0 - 2 point, where 1 goal = 1 point.
2. Technical documentation: 0 - 6 points, including:
- a. determination of the technological stack consisting of at least three elements: 0 - 3 points, where 1 element = 1 point,
 - b. database documentation or diagram (overall or for a selected part of the application): 0 - 3 points, where 1 table + marking in it primary and foreign keys and indexes (if any) = 1 point
3. Functional layer: 0 - 20 points, including:
- a. compliance of the implementation with the functional requirements: 0 - 10 points (1 requirement / 2.5 points),
 - b. compliance of the implemented architecture with the standards of the selected framework or design pattern, with particular emphasis on the separation of application layers: 0 - 10 points.
4. Use of the database (relational or non-relational): 0 - 20 points, including:
- a. (relational database) normalized structure of the database (at least the second normal form): 0 - 5 points
or
(non-relational database) criteria for 5 points in this respect depend on the choice of non-relational database technology - please contact me if you choose this option
 - b. compliance of the database structure and implementation in the application with the specification: 0 - 5 points,

- c. the use of at least three of the four core CRUD operations: 0 – 10 points, where 1 type of operation = 3.5 points.
5. View layer: 0 – 20 points, including:
- a. compliance of the view layer with selected parts of the specification: 0 – 5 points,
 - b. guaranteeing correct display regardless of the device size for at least one screen: 0 – 15 points, where correctness on 1 screen for three different, standard sizes = max 10 points (1 size / 5 points).
6. Using naming conventions in programming and formatting the code typical for the selected technology: 0 – 10 points.
7. Demonstration of the specification and the source code of the project along with an indication of the above-mentioned elements: 0 – 10 points; alternatively, the demonstration can be replaced with multimedia material or extended documentation, showing the compliance of the implemented project with the points indicated above.